

Assignment

Ethical, Social, and Legal Implications of AI Systems



Inspiring Excellence

Department of Computer Science and Engineering

Student Name: Avishek Biswas

Course: CSE-422

Student ID: 23201427

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Case 7: Pushing Sales During “Prime Vulnerability Moments”

Question 1: Environmental and Societal Impact

Impact on the Society

The proposed system deliberately exploits periods of emotional fragility by identifying prime vulnerability moments to increase purchases of consumers. One major issue is how this might weaken a person's ability to think for themselves and make their own decisions. For example, the AI might target women on Monday if it knows by historical stats that a certain group of women feel less confident about their body on Mondays, which is a misuse of technology . If these business systems become common, it might lead to a world where being manipulated by an algorithm is considered normal. Long term deployment of these systems normalizes algorithmic manipulation and removes human autonomy by restricting meaningful opportunities for conscious choice. Academic research indicates that when platforms leverage superior behavioral data for manipulation, consumer welfare predictably decreases while corporate profits artificially inflate (Acemoglu et al., 2023).

Impact on the Environment

By collecting and analyzing data 24/7 to find small emotional changes is harmful for the environment. A huge amount of electrical and computational power is required to keep these big AI models running 24/7. To keep these massive data centers cool also, a huge amount of water is required which hampers local water supplies moreover it is environmentally unsustainable . According to Bolthouse(2025), cooling these massive data centers heavily impacts local ecologies, with projected water withdrawals for AI operations reaching billions of gallons annually.

Question 2: Ethical Violations Regarding Bias, Transparency, and Accountability

Bias

The way this system works seems to rely heavily on stereotypes. It is built and trained on the idea that certain groups of people act or feel in a specific way for a specific period of time just because of their gender or the day of the week. By using these negative assumptions about body image, the system might be guilty of unfair emotional profiling.

Transparency

There is a serious lack of transparency in this case which is a major ethical issue. The AI is used to limit or change what a person thinks of their options, but it does this without the user ever knowing. Because the system works in the background, people are being watched and influenced without their awareness. It is difficult for anyone to give informed consent" when they do not even realize their mood is being tracked.

Accountability

It often feels like there is no real way to hold anyone responsible for these actions. Companies might hide what they are doing by using filler words like empowering the customer. It seems that by letting the math do the work, firms can avoid being blamed for exploiting people. According to Petropoulos(2022) , the success of manipulative AI strategies mostly depends on algorithms, allowing firms to extract profit from vulnerable consumers while evading public scrutiny.

Question 3: Legal Implications and Engineering Responsibilities

Legal Implications

If a company uses these tricks, they can face serious and legal trouble. New rules are being made everyday to stop this kind of exploitation marketing and usage of loopholes. The EU AI Act explicitly prohibits AI practices using subliminal, purposefully manipulative, or deceptive techniques (Future of Privacy Forum, 2026). Additionally, figuring out how someone feels is a very private matter. Deducing a person's emotional state qualifies as processing mental data, and doing so without explicit, informed consent directly violates the General Data Protection Regulation (GDPR) (Ienca & Malgieri, 2022).

Engineering Responsibilities

The engineers have a duty to follow the law and ethics and follow them accordingly. This means they should probably refuse to write code/ make algorithms that are specifically meant to exploit people's feelings just for making profits. It is essential for the engineers and architects who are building these systems, to make sure that humans always remain in control of the situation. They must ensure that AI is not used to trick the human brain. To keep things fair, engineers could also let supervisors/ experts check their work. By allowing independent organizations to strictly inspect the system, it is more likely that hidden social problems will be pointed out before the AI is ever used.

References

1. Acemoglu, D., Makhdoumi, A., Malekian, A., & Ozdaglar, A. (2023). A model of behavioral manipulation (NBER Working Paper No. 31872). National Bureau of Economic Research. <https://ideas.repec.org/p/nbr/nberwo/31872.html>
2. Bolthouse, J. (2024). Data drain: The land and water impacts of the AI boom. Land Lines. Lincoln Institute of Land Policy. <https://www.lincolninst.edu/publications/land-lines-magazine/articles/land-water-impacts-data-centers/>
3. Future of Privacy Forum. (2026). Red lines under the EU AI Act: Understanding manipulative techniques and the exploitation of vulnerabilities. Future of Privacy Forum. <https://fpf.org/reports/eu-ai-act-red-lines-manipulation>
4. Ienca, M., & Malgieri, G. (2022). Mental data protection and the GDPR. Journal of Law and the Biosciences, 9(1), lsab036. <https://doi.org/10.1093/jlb/lsab036>
5. Petropoulos, G. (2022). The dark side of artificial intelligence: Manipulation of human behaviour. Bruegel. <https://www.bruegel.org/blog-post/dark-side-artificial-intelligence-manipulation-human-behaviour>
6. Stahl, B. C., Schroeder, D., & Rodrigues, R. (2022). *Ethics of Artificial Intelligence: Case Studies and Options for Addressing Ethical Challenges*. Springer International Publishing.